

Level 13: EAS Build Pipeline, Native Credentials & App Store Submissions

React Native Masterclass

Expo SDK 54 / RN 0.81

Production CI/CD

13.1 — EAS Build Cloud Architecture

To compile native iOS binaries (`.ipa`), Apple strictly requires an Apple Mac running Xcode. **EAS Build** allows developers on Windows laptops to compile iOS and Android apps by offloading native builds to Expo's managed cloud server clusters in ~8 minutes.

Tenglish Explanation:

Windows laptop lo Xcode undadu kabatti iPhone apps direct ga compile avvavu. EAS Build mana code ni cloud లో real Apple Mac servers లో compile chesi ready-to-use `.ipa`` and `.aab`` links ichesthundi.

13.2 — Android App Bundles (.aab) & Keystore Signing

.apk vs .aab: Legacy `.apk` contains binaries for all CPU architectures (~50MB). Modern `.aab` allows Google Play to generate optimized, tiny 15MB downloads specifically tailored to each user's phone processor.

Android Keystores: Cryptographic security certificate (`.keystore`) used to digitally sign app updates. Without the original keystore, Google Play blocks future updates. EAS Credentials securely encrypts and persists keystores in the cloud.

Tenglish Explanation:

Play Store lo AAB format vaadithe app size 50% tagguthundi. Keystore ante mee app యొక్క digital signature stamp; idi unte ne Play Store kotha updates (v2.0) accept chesthundi.

13.3 — iOS Certificates & Provisioning Profiles

Security Asset	Format	Role in iOS Ecosystem
Bundle Identifier	<code>com.org.app</code>	Unique global identity registered in Apple Developer Portal.
Distribution Certificate	<code>.p12</code>	Digital developer passport proving legal identity.
Provisioning Profile	<code>.mobileprovision</code>	Security permit linking Bundle ID, certificate, and target devices.

Tenglish Explanation:

iPhone lo app install avalante 3 things kavali: 1. Bundle ID (Unique name), 2. Certificate (Developer passport), 3. Provisioning Profile (Security permit). EAS okka command tho ee 3 files ni Apple nunchi automatic ga configure chesthundi.

13.4 — Multi-Environment Build Profiles & EAS Submit

Professional applications isolate environments using `eas.json` and store API secrets securely in Expo's encrypted cloud vault:

```
{
  "build": {
    "development": { "developmentClient": true, "distribution": "internal" },
    "preview": { "distribution": "internal", "android": { "buildType": "apk" } },
    "production": { "android": { "buildType": "app-bundle" } }
  }
}
```

EAS Submit: Automatically transfers compiled `.aab` and `.ipa` binaries directly into Google Play Console and Apple App Store / TestFlight via `eas submit --platform all`.

Tenglish Summary:

`eas.json` lo Dev (Local), Staging (Test APK), and Production (Real App Store) environments isolate chestham. `eas submit` command terminal nunchi direct ga Google Play & Apple TestFlight loki apps ni 2 mins lo upload chesthundi!